Stress De-stress
MINI PROJECT SYNOPSIS

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

By

Anand Misra	Pratham Nahata
04011502723	05611502723
Neha Kamal	Mehak
01111502723	06311502723

Guided by

Dr. Jolly Parikh

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BHARATI VIDYAPEETH'S COLLEGE OF ENGINEERING

(AFFILIATED TO GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY, DELHI)

SEPTEMBER 2025

INTRODUCTION

The pervasive nature of stress in contemporary society poses a significant threat to mental health, productivity, and overall well-being. Individuals across various demographics, from students to professionals, are grappling with the effects of prolonged stress, which can lead to mental fatigue, burnout, anxiety, and depression. While a multitude of digital wellness applications and traditional stress management techniques exist, they often fall short in providing timely, personalized, and engaging interventions that can adapt to the dynamic needs of the user [1, 2, 3]. A key limitation of existing approaches is their passive, one-size-fits-all model, which fails to provide the continuous, real-time support required for effective stress management. This gap highlights a critical need for a more sophisticated, proactive, and individualized approach to mental wellness [4, 5].

This project addresses these limitations by proposing an AI-enabled framework that integrates real-time stress detection with personalized, gamified intervention mechanisms. Our system is designed to move beyond the static model of current digital health tools by leveraging cutting-edge AI technologies to create a dynamic and adaptive platform for mental wellness. A key component of this framework is the use of an Online Reinforcement Learning-based Retrieval-Augmented Generation (OnRL-RAG) system [6], which allows the model to continuously learn from user feedback and adapt its responses in real time. This dynamic learning process ensures that the system provides highly personalized and contextually relevant support, addressing a critical need identified in a review of existing mental health support technologies [7]. Other adaptive systems, such as SouLLMate [8], have also demonstrated the potential for LLM-driven applications to offer advanced features like risk detection and proactive guidance.

The framework will utilize computer vision techniques to analyze physiological and behavioral indicators, such as facial expressions and micro-gestures, to accurately detect and classify a user's stress level in real time [9]. This non-invasive monitoring is the foundation for the system's ability to provide timely interventions. The stress detection module draws inspiration from recent advancements in the field, including the use of deep learning models that analyze various physiological signals like blood volume pulse (BVP) [10], electroencephalogram (EEG) [11], and multimodal data (e.g., electrocardiograms, electrodermal activity, and facial expressions) [12, 13, 16]. The system is designed to be invariant to individual differences in stress responses, a challenge that has been addressed by domain generalization approaches [14].

Once stress is detected, the system's recommendation module will suggest

personalized interventions from a range of evidence-based techniques, including guided breathing exercises, mindfulness meditation, and adaptive soundscapes. A review of sound-based interventions has shown that specific auditory stimuli, such as music, can effectively reduce physiological stress markers like cortisol levels and heart rate variability [15]. By personalizing these interventions, the system ensures they are both effective and engaging.

Furthermore, a unique aspect of this project is the integration of gamification to enhance user engagement. By transforming traditional coping strategies into interactive and enjoyable activities, the system aims to increase user participation and long-term adherence to stress-relief practices. This dual capability—real-time stress detection and interactive, personalized relief—positions the system as an intelligent, empathetic assistant rather than a static tool. The project's ultimate goal is to bridge the gap between traditional wisdom and modern digital tools, providing a comprehensive and effective solution for proactive stress management.

LITERATURE REVIEW

Title	Year & Citation	Technology Used	Dataset	Key Findings	Results	Limitations	Future Scope
Stress Recognition from Multimodal Physiological Signals	2025, Yang et al.	Deep Learning (CNN, GASF, GADF, MTF)	ECG, HRV, GSR signals from controlled experiments	Transforming signals into image representations enhances stress classification	Effective stress detection	Lab environment, small sample size	Real-world, mobile deployment
SouLLMate: Adaptive LLM for Mental Health	2024, Guo et al.	LLM, RAG, Chain-of-Thought	Conversational datasets from online counseling sessions	LLM provides personalized mental health guidance	Adaptive support	Not validated in real-world settings	Integrate with physiological sensors and mobile apps
OnRL-RAG: Real-Time Personalized Mental Health Dialogue	2025, Bilal et al.	LLM + RAG + RLHF	Annotated user dialogues	Combines retrieval and generative models for tailored interactions	Real-time adaptive responses	Requires extensive annotated data; no physiological signals	Hybrid ML + LLM integration for stress monitoring
Smartwatch-based Stress Detection	2025, Fried et al.	Wearables (Garmin Vivosmart 4)	Heart rate & activity records from volunteers	Smartwatches often misclassify stress versus physical exertion	Limited reliability	Signal noise, device constraints	Combine with ML models for accurate stress detection
Never Alone App	2025, AIIMS Delhi	Mobile App, Basic ML	Student survey responses	Provides support and coping resources for students	Increased engagement	Not dynamically personalized	Integrate real-time stress monitoring and LLM suggestions
Large Language Models for	2025, Nori et al.	LLMs, Domain Fine-tuni	Clinical textual records	Generic LLMs perform	Needs adaptation	Restricted to textual	Adapt for mental health

Clinical Summaries		ng		poorly without domain adaptation		data; not real-time	recommendation systems
RAG & Hybrid Models in Mental Health	2025, Amugongo et al.	RAG, GraphRAG, Hyper-RAG	Clinical and counseling datasets	Enhances factual grounding and recommendation quality	More consistent outputs	Limited patient-specific testing	Integrate with stress detection apps for real-time guidance
Mental Health Apps & LLM-based Recommendations	2025, Zhang et al.	LLM, Mobile Applications	User-generated entries and survey feedback	Context-aware recommendation generation	Improved user engagement	No integration with physiological signals	Hybrid ML + LLM system for real-time personalized support
Nature Study on Stress & ML	2025, s41598-025-01228-3	ML/DL + LLM	Combined physiological measurements and self-reports	Detection and intervention recommendations combined	Promising results	Conducted in lab setting	Extend to mobile real-time applications
Nature Study: Stress & Coping Patterns	2025, s41598-025-06918-6	ML + Survey analysis	Behavioral surveys with physiological annotations	Identifies stress patterns and coping mechanisms	Provides actionable insights	Limited diversity in sample	Integrate into personalized mental health apps
Frontiers in Computer Science: Mental Health LLMs	2025, Frontiers	LLM + DL	Textual data from counseling sessions	LLMs can explain stress patterns in user-friendly language	Improved interpretability	Limited user testing	Integrate into real-time recommendation apps
JMIR Mental Health Study	2025, e69120	ML + LLM	Survey responses with physiological signals	Generates personalized mental health recommendations	User-centric support	No real-time feedback	Integrate into mobile app platforms

SouLLMate: Enhancing Diverse Mental Health Support with Adaptive LLMs, RAG and Prompt Engineering.	2024	LLM + RAG	Text corpora from mental health forums	Retrieval-augmented generation enhances factual recommendations	More contextually relevant suggestions	No physiological data	Hybrid ML + LLM stress detection apps
OnRL-RAG: Real-Time Personalized Mental Health Dialogue System	2025	LLM + Chain-of-Thought	Counseling conversation transcripts	Stepwise reasoning improves recommendation accuracy	More coherent outputs	Not tested in real users	Real-time adaptive recommendation systems
TEANet: A Transpose-Enhanced Autoencoder Network for Wearable Stress Monitoring	2025	ML + LLM	Survey data combined with physiological metrics	Multi-input integration for personalized guidance	Feasible recommendations	Limited dataset	Mobile app deployment for real-time guidance
Brain2Vec: EEG-Based Stress Detection using Deep Learning Framework	2025	LLM + RLHF	User conversation datasets	RLHF improves adaptation to user responses	Adaptive recommendations	Needs large annotated datasets	Combine with real-time stress sensors for hybrid system

PROBLEM STATEMENT

Despite the widespread availability of digital wellness applications, a significant gap exists in providing genuinely personalized and real-time stress management support. Existing systems are often static and lack the ability to dynamically adapt to an individual's fluctuating emotional state and personal context. This project addresses this deficiency by developing an AI-integrated framework that leverages an Online Reinforcement Learning-based Retrieval-Augmented Generation (OnRL-RAG) system to provide a truly adaptive, proactive, and personalized solution for stress detection and intervention.

OBJECTIVES

- To identify and analyze key physiological and behavioral indicators that can effectively reflect stress levels.
- To collect, preprocess, and structure stress-related datasets for model training and evaluation.
- To develop and train a machine learning/deep learning model for accurate stress detection and classification.
- To design and integrate a recommendation module that suggests personalized interventions (e.g., music, meditation, breathing exercises).
- To build a mobile/desktop application that provides real-time stress monitoring and recommendations through a user-friendly interface.
- To incorporate user feedback into the system for continuous improvement of detection accuracy and recommendation quality.
- To evaluate the application using technical performance metrics (accuracy, precision, recall, F1-score) and user-centric measures (satisfaction, usability)

METHODOLOGY

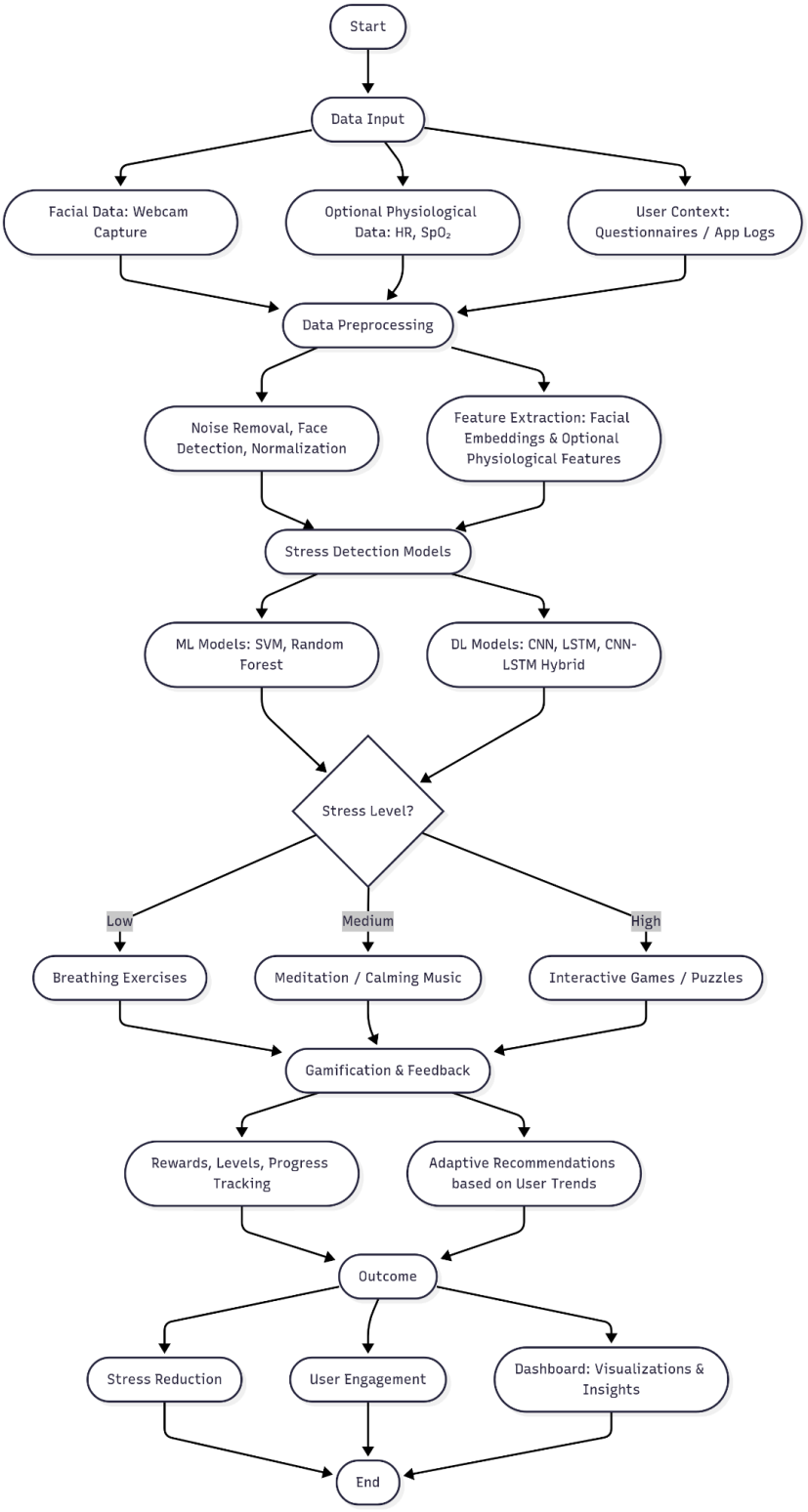

Figure 1: Proposed Methodology Diagram

The proposed work will be carried out in multiple phases to ensure systematic development, evaluation, and deployment of a reliable stress detection and recommendation system with an application interface.

1. Data Collection & Preparation

- **Data Sources**: Collect facial expression datasets (e.g., FER-2013, AffectNet, or custom webcam captures) and stress-related text/audio datasets if needed.
- **Preprocessing**:
  - Normalize images (resizing, grayscale/augmentation for robustness).
  - Apply noise removal and face landmark detection (using OpenCV/Mediapipe).
  - Label stress levels (low, medium, high).
- **Optional Physiological Signals**: If IoT sensors are used (heart rate, SpO_2), synchronize and preprocess readings.

2. Feature Extraction & Detection Strategy

- **Facial Analysis**:
  - Detects facial landmarks and micro-expressions using Mediapipe/dlib.
  - Extract emotion-based features relevant to stress.
- **Signal Processing (Optional)**:
  - Use heart-rate variability or oxygen saturation to complement visual stress cues.
- **Fusion Approach**: Combine multimodal inputs (facial + physiological) for improved accuracy.

3. Stress Classification Models

- **Baseline Models**: Train classical ML models (SVM, Random Forest, Logistic Regression).
- **Deep Learning Models**: Implement CNN, LSTM, or hybrid CNN-LSTM for temporal sequence stress detection.
- **Evaluation**: Compare models using F1-score, recall, and confusion matrices for low/medium/high stress classification.

4. Recommendation & Intervention Module

- **Personalized Suggestions**:
  - Low Stress → Motivational tips, focus reminders.
  - Medium Stress → Guided breathing exercises, short breaks.
  - High Stress → Meditation tracks, calming music, interactive relaxation games.
- **Gamification**: Integrate simple games (puzzles, breathing guides, reflex games) as active stress relief interventions.

5. Application Development & Integration

- **Frontend**: Build user dashboard (ReactJS / Streamlit) for real-time visualization.
- **Backend**: Use Flask/Django for model integration and API serving.
- **Features**:
  - Live webcam stress detection.
  - Dashboard with real-time stress levels and history.
  - Recommendations and activity triggers.
- **User Insights**: Track trends over time and provide feedback loops for long-term

stress management.

6. Validation & Performance Evaluation

- **Quantitative Metrics**: Precision, Recall, F1-Score, Stress detection rate.
- **User Testing**: Pilot test with participants for feedback on usability and effectiveness.
- **Comparative Analysis**: Benchmark performance of ML vs. DL models.

7. Deployment & Future Enhancement

- Package as a **desktop/web/mobile application** for accessibility.
- Future extensions:
  - Integration with wearables (Fitbit, smartwatches).
  - Incorporation of LLMs for **personalized coping recommendations**.
  - Expansion to multi-modal (speech + text + facial cues).

EXPECTED OUTCOME

1. Proactive & Adaptive AI System

Develop a real-time, personalized AI system for stress detection.
Utilize OnRL-RAG for continuous adaptation based on user feedback.

2. Dynamic & Engaging Interventions

Implement a recommendation module for tailored, real-time interventions.
Integrate gamification to boost user motivation and adherence.

3. Actionable Behavioral Insights

Generate insights on individual stress triggers and patterns.

Provide data-driven recommendations for proactive self-management.

4. Contribution to AI & Mental Health Domain

Demonstrate the OnRL-RAG framework as a scalable solution for mental wellness.
Establish a novel model for empathetic, effective, and continuous digital therapeutics.

5. Comprehensive Project Validation

Evaluate performance with technical and user-centric metrics.
Validate the system's effectiveness in providing tangible stress relief and improving well-being.

APPLICATION

1. Digital Therapeutics

Provides accessible, personalized, and stigma-free mental health support.
Acts as a proactive, conversational tool for managing stress, anxiety, and depression.

2. Corporate & Educational Wellness

Integrates into workplace programs to mitigate burnout and enhance productivity.
Supports students during high-pressure periods to improve academic performance.

3. Clinical Decision Support

Serves as an assistive tool for psychologists and counselors.
Provides real-time, data-driven insights into a patient's emotional state to accelerate communication efficiency.

4. Personal Wellness & Gamification

Offers an AI-driven, gamified wellness app for individual use.
Transforms passive coping strategies into interactive and engaging activities.

5. Advanced Mental Health Research

Provides a dynamic platform for collecting rich data on stress patterns and intervention effectiveness.
Contributes to sociological, psychological, and neuroscientific research by aligning theories with real-world human behavior

REFERENCES

[1] K. Oh *et al.*, "Supervised Contrastive Learning-Based Stress Detection for Sensor-Based Stress Monitoring," *Elsevier*, 2025.

[2] A. Pinge *et al.*, "Detection and Monitoring of Stress Using Wearables: A Systematic Review," *Frontiers in Computer Science*, vol. 6, 2024.

[3] S. Gedam *et al.*, "Analyzing Mental Stress in Indian Students through Shuffled ECA-Net," *Scientific Reports*, 2025.

[4] M. S. Ali, S. S. Bipro, M. E. H. Chowdhury, *et al.*, "TEANet: A Transpose-Enhanced Autoencoder Network for Wearable Stress Monitoring," *arXiv preprint arXiv:2503.12657*, 2025.

[5] M. Mynoddin, T. Dev, and R. Chakma, "Brain2Vec: A Deep Learning Framework for EEG-Based Stress Detection Using CNN-LSTM-Attention," *arXiv preprint arXiv:2506.11179*, 2025.

[6] E. Oliver and S. Dakshit, "Cross-Modality Investigation on WESAD Stress Classification," *arXiv preprint arXiv:2502.18733*, 2025.

[7] Y. Xiao, H. Sharma, S. Kaur, *et al.*, "Human Heterogeneity Invariant Stress Sensing (HHISS)," *arXiv preprint arXiv:2506.02256*, 2025.

[8] M. Saskovets *et al.*, "Effects of Sound Interventions on the Mental Stress Response in Adults: Scoping Review," *JMIR Mental Health*, 2025.

[9] Y. Shi *et al.*, "An AI-Driven Sound Therapy System Design for Enhancing Mental Health of Older Adults," in *Lecture Notes in Computer Science (LNCS)*, ACM, 2024.

[10] "Automated Video and Audio Based Stress Detection using Deep Learning Techniques." [Online]. Available: https://share.google/adpfPJwiqb4xD52R0

[11] Y. Jin *et al.*, "The Applications of Large Language Models in Mental Health: Scoping Review," *Journal of Medical Internet Research*, vol. 27, May 2025, doi: 10.2196/69284.

[12] S. Yang *et al.*, "A deep learning approach to stress recognition through multimodal physiological signal image transformation," *Scientific Reports*, vol. 15, no. 1, Jul. 2025, doi: 10.1038/s41598-025-01228-3.

[13] Q. Guo *et al.*, “SouLLMate: An Application Enhancing Diverse Mental Health Support with Adaptive LLMs, Prompt Engineering, and RAG Techniques,” *arXiv preprint arXiv:2410.16322*, Oct. 2024, doi: 10.48550/arXiv.2410.16322.

[14] “Chapter reference,” in *Proceedings of the International Conference on Human-Computer Interaction*, Springer, 2024. [Online]. Available: https://dl.acm.org/doi/10.1007/978-3-031-60615-1_30

[15] “arXiv preprint: Stress detection study,” Apr. 2025. [Online]. Available: https://arxiv.org/html/2504.02894v1

[16] “Research article on stress recognition,” *Expert Systems with Applications*, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417425008139